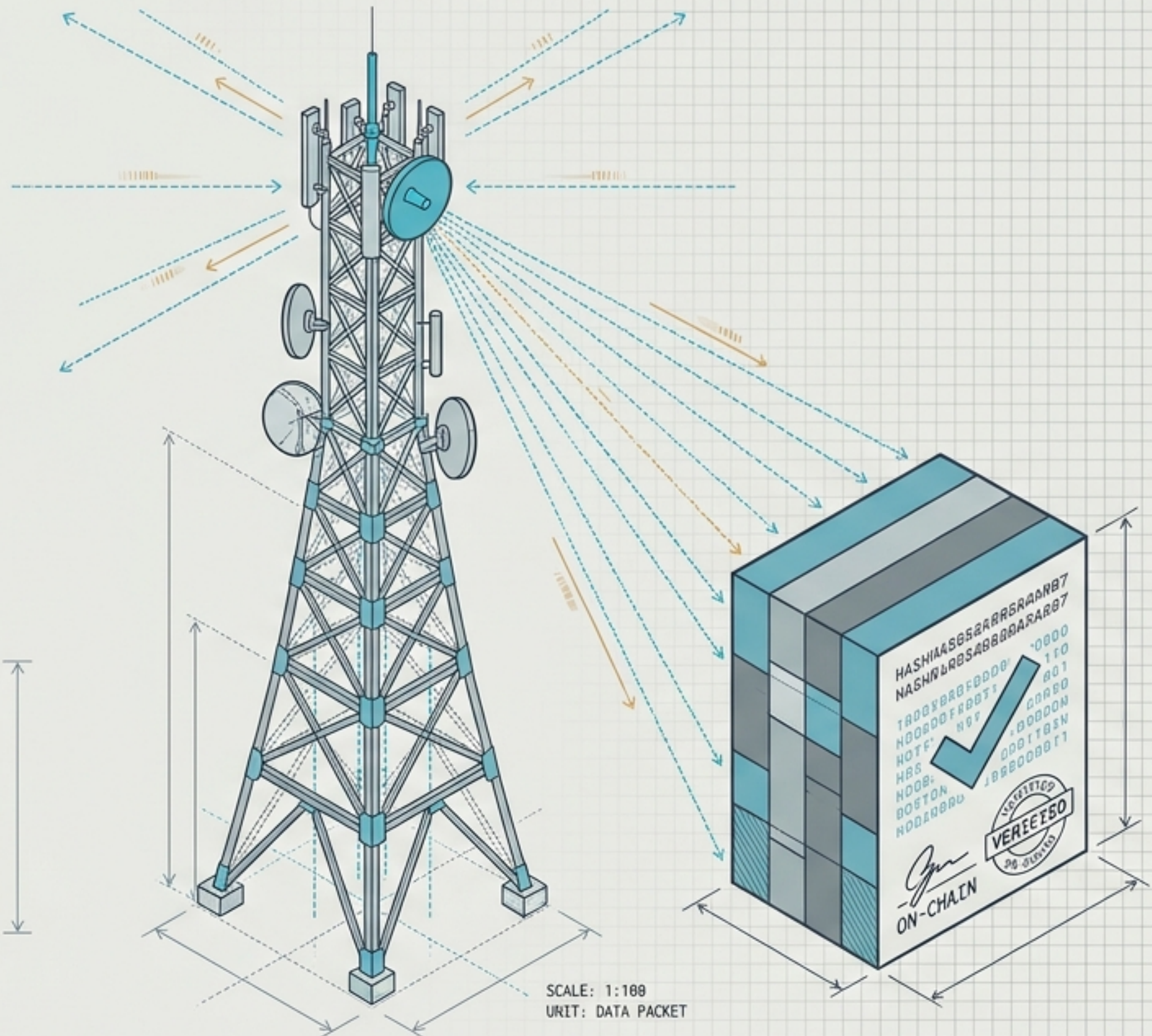


The Blueprint for Verified Broadcast

A diagram of a terminal window. The title bar at the top reads "TERMINAL OUTPUT // DATA STREAM" on the left and "REV 3.14" on the right. The main content area displays the following text:

```
INFRASTRUCTURE: EchoStar 5,800-Node DePIN
PROTOCOL: Media over QUIC (MoQ)
VERIFICATION: On-Chain Proof-of-Delivery (Base L2)
YIELD: $45-$65 FAST Premium CPMs
```

The terminal window is surrounded by dimension lines: a horizontal double-headed arrow at the top, a vertical double-headed arrow on the left, and a vertical double-headed arrow on the right.



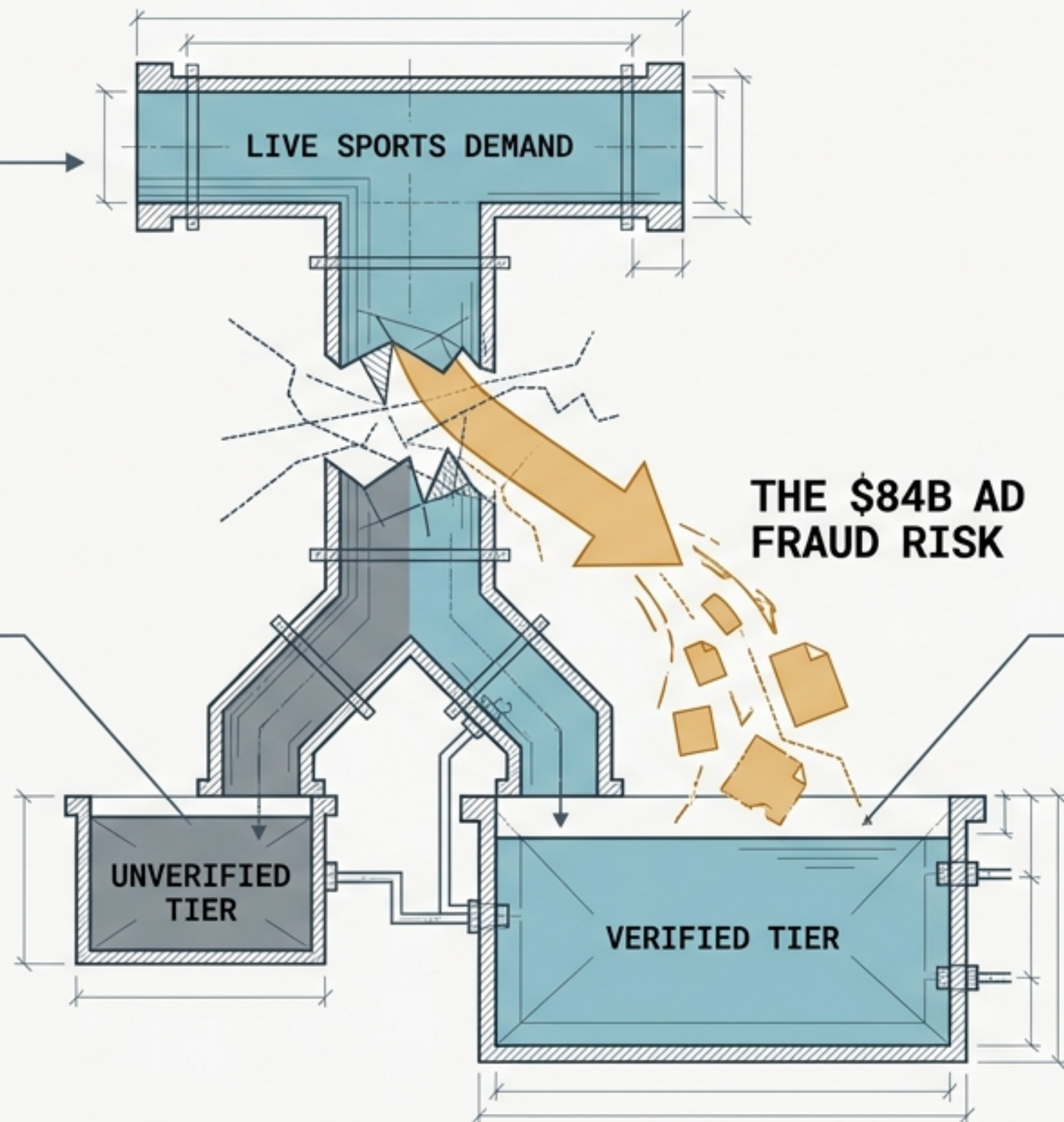
THE STRUCTURAL DEFICIT IN LIVE SPORTS STREAMING

Supply-Demand Imbalance:

Viewers spend 37.5% more on sports streaming, yet FAST fill rates remain low due to unverified delivery.

The Unverified Trap: Live streams reach 200+ markets but are restricted to \$15-\$25 CPMs on private marketplaces.

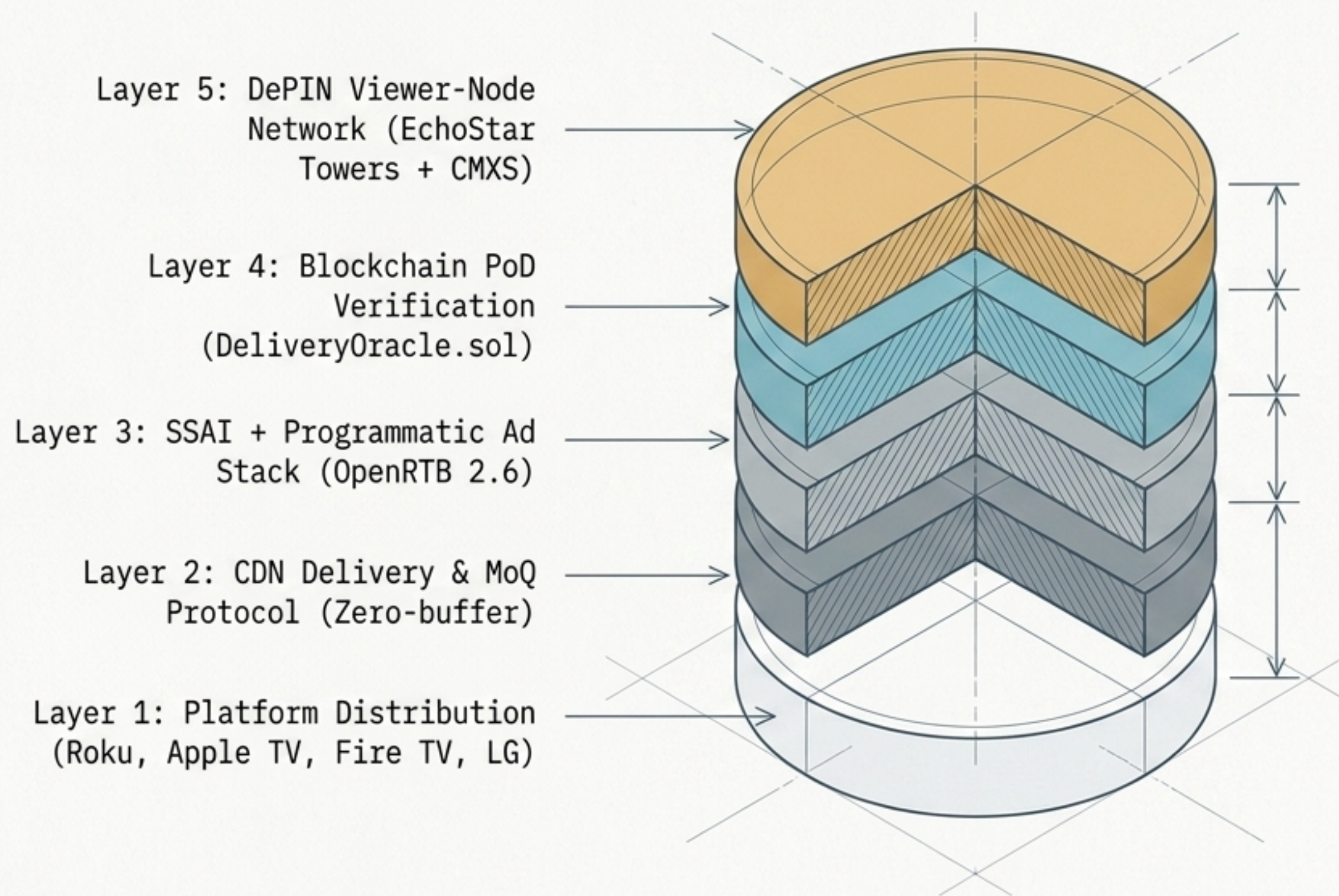
Third-party validation cannot operate at live-stream latency.



**THE \$84B AD
FRAUD RISK**

The Verified Premium: Cryptographically proving a real human watched an ad unlocks \$45-\$65+ CPMs.

The CMXS Full-Stack Architecture



Integration as a Structural Moat

Traditional FAST operators lack verification. Verification vendors lack physical edge nodes. CMXS integrates physical towers, transport protocol, and token economics into a single closed-loop machine.

The Physical Foundation: 5,800 Edge Nodes

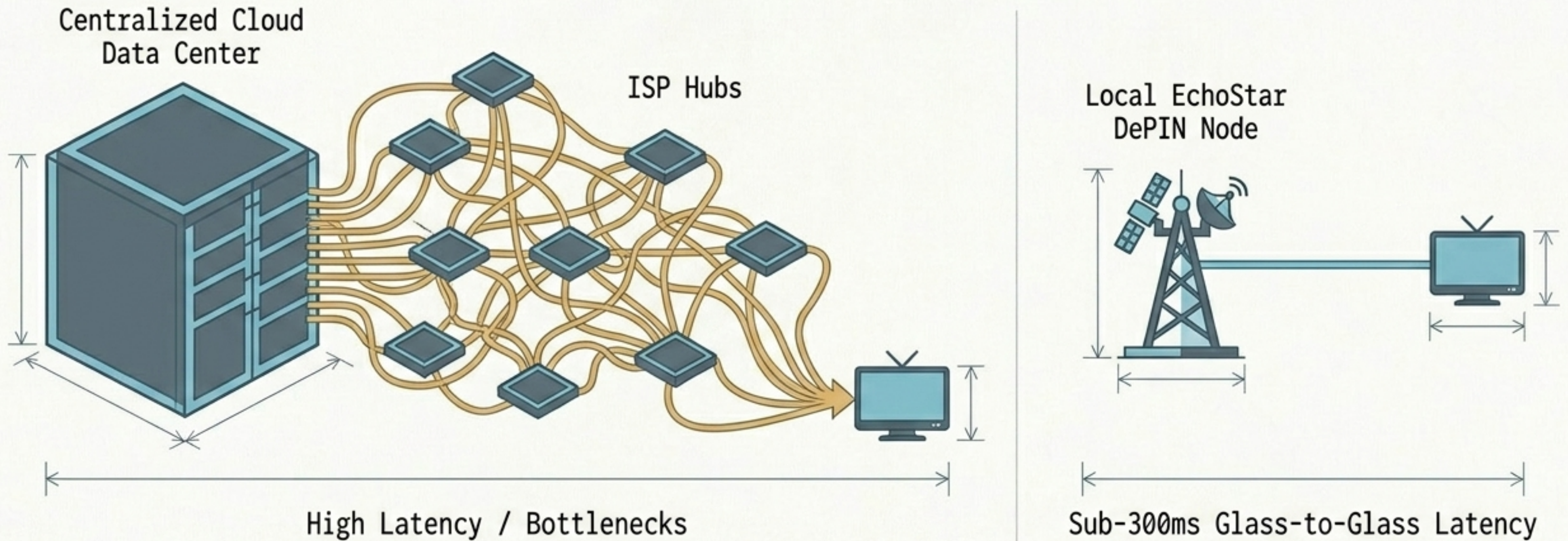


Zero Cold-Start Problem
Unlike typical DePIN networks that must bootstrap hardware for years, CMXS launches with an immediate, active nationwide physical footprint.

Last-Mile Delivery
Towers function as hyper-local delivery nodes via SRT receive paths, bypassing congested centralized public clouds.

SLA Enforcement
EchoStar towers stake CMXS tokens as performance collateral. Slashed stakes from underperformance flow directly to the protocol treasury.

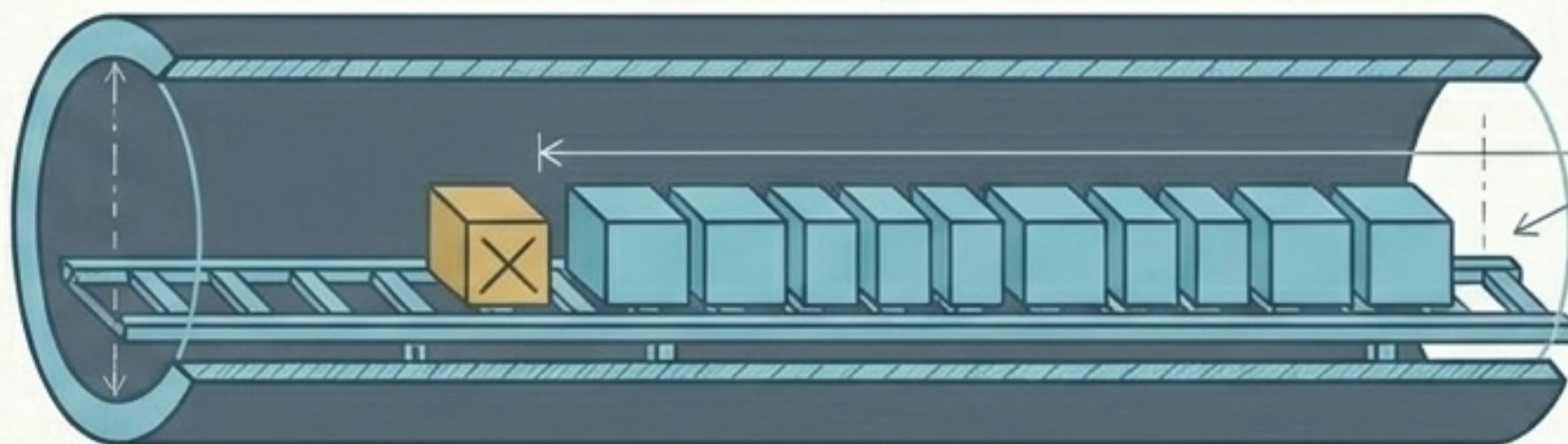
The Last-Mile Physical Advantage



A local TV station in a small DMA receives the exact same sub-second delivery latency as New York. The nearest physical EchoStar tower handles the final hop-not the nearest major cloud data center.

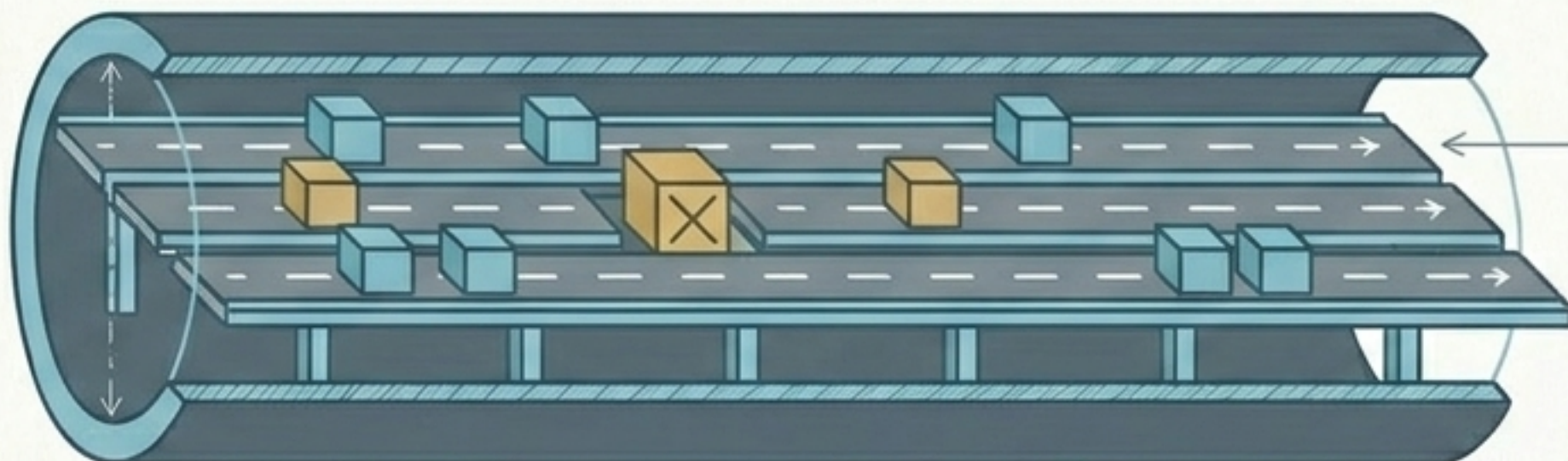
Eliminating Transport Bottlenecks: MoQ vs. HLS

HLS over TCP



Head-of-line blocking:
A single dropped packet stalls the entire video frame.

MoQ over QUIC



Parallel delivery: Video objects travel independently. Dropped packets do not block parallel frames.

← The WebTransport API over UDP completely eliminates transport-level blocking. →
← Result: Millions of simultaneous viewers receive live sports without buffering. →

Delivery Protocol Matrix

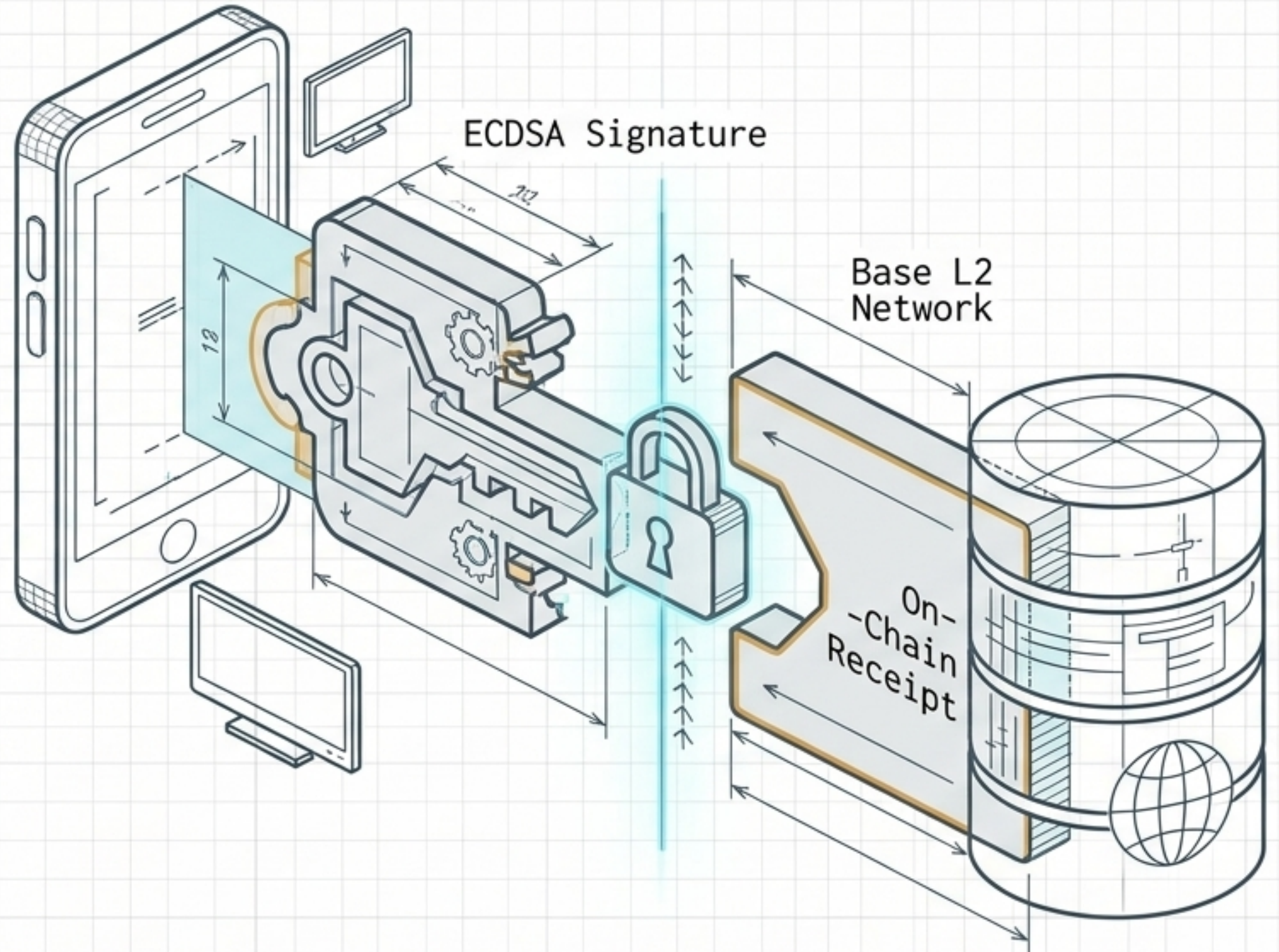
Protocol	Glass-to-Glass Latency	Scale Ceiling	Ad Insertion
Traditional HLS FAST	10-30s Latency	Unlimited Scale	SSAI Stitching
LL-HLS	2-5s Latency	Unlimited Scale	Limited SSAI
WebRTC	100-500ms Latency	~1,000 concurrent max	No Ad Insertion
CMXS MoQ (Caton)	200-300ms Latency	Millions concurrent	SGAI + SSAI

Takeaway: MoQ is the only protocol capable of pairing WebRTC-like interactivity with CDN-scale fan-out.

The Cryptographic Handshake: Proof-of-Delivery

1. **100% Ad Completion:** CMXS SDK autonomously generates a signature combining Edge Node ID, Impression ID, and Unix Timestamp.
2. **Oracle Verification:** Oracle checks fraud heuristics and verifies the device signature against the registered wallet.
3. **Base L2 Minting:** DeliveryOracle.sol mints an immutable on-chain receipt within 3 seconds.

The signature is cryptographically unique to the device and moment. It cannot be spoofed, fabricated, or replicated by bot farms.

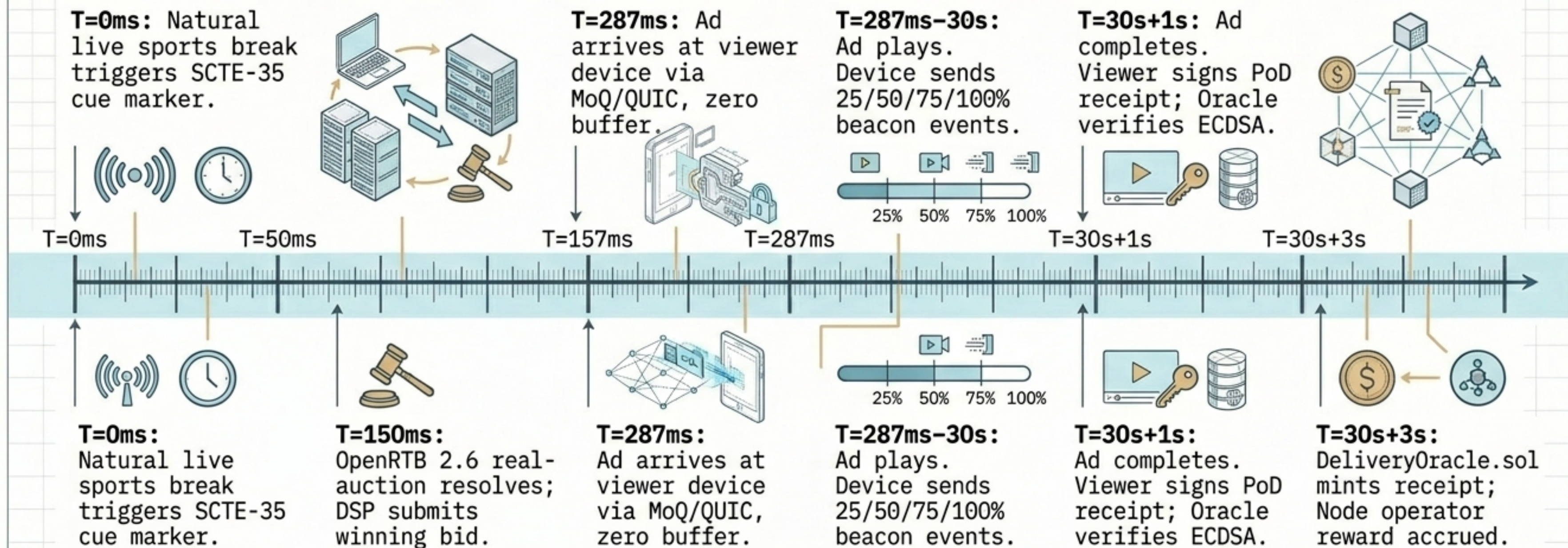


The Verification Premium: Monetization Reality Check

Standard FAST (Unverified)	Third-Party (DoubleVerify/IAS)	CMXS PoD (On-Chain)
CPM Range: \$10-\$25	CPM Range: \$20-\$35	CPM Range: \$45-\$65+
Fraud Risk: High (1 in 5 impressions fake)	Fraud Risk: Medium	Fraud Risk: Negligible
Audit Trail: None	Audit Trail: Partial / Proprietary	Audit Trail: Immutable, Public (Basescan)

Advertisers are not unwilling to pay premium rates for sports;
they are unwilling to pay premium rates for unverified inventory.
On-chain validation eliminates the trust deficit.

The 35-Second Economic Engine



**Total verified impression cycle: under 35 seconds.
Fully automated. No human in the loop.**

DePIN Tokenomics: The Burn-Mint Equilibrium

Fiat Inflow

Advertisers pay USDC to purchase verified impressions.



EchoStar

New Mint



Viewer Node

Verified Impression

Node Reward (Mint)

EchoStar towers and viewer nodes earn newly minted CMXS for delivering the verified impression.



Verified Node

Structurally tied to corporate ad spend, not speculation.

Mechanical Burn

AdBurn.sol automatically burns CMXS tokens proportional to ad spend.



AdBurn.sol



Burn Rate: Incinerator

Burn Rate: Proportional

Circulating Supply



Deflationary Pressure

Total circulating supply decreases mechanically as corporate ad spend increases.

Ad Spend

Decrease

Hardware Participation: DePIN Reward Tiers

Tier 3 (3.0x): EchoStar CBRs Radio Node

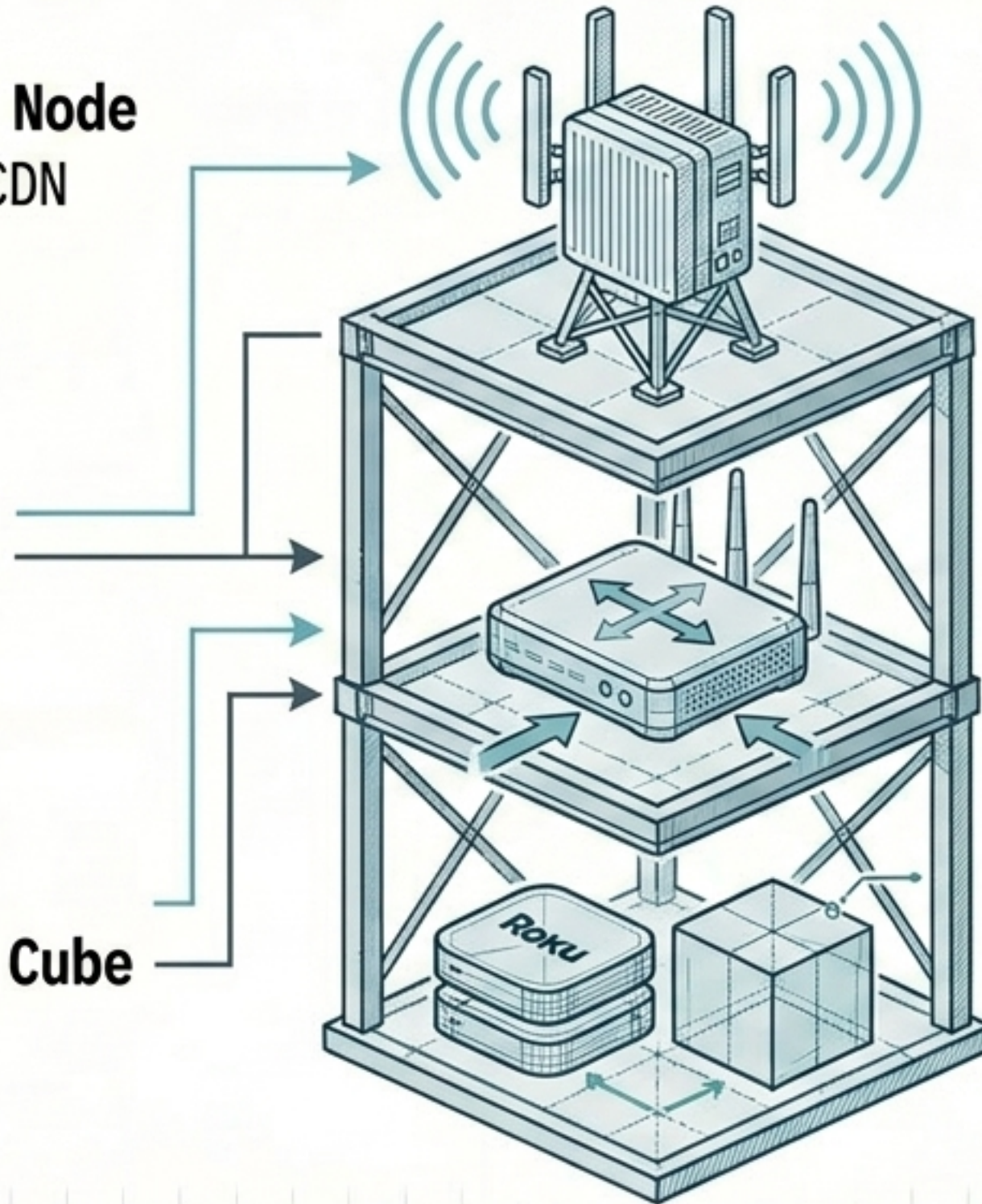
Wireless coverage + CDN
small cell router.

Tier 2 (1.5x): Home Broadband Node

Full CDN caching,
high uplink.

Tier 1 (1.0x): Roku Ultra / Fire TV Cube

IP data relay &
CDN caching.



The Math

- Base Reward: 0.001 CMXS per verified ad impression delivery.
- At 500,000 monthly viewers = 2,250 CMXS/month per channel in node rewards.
- Nodes are dynamically promoted to active caching status based on uptime and SLA compliance.

Local TV Go-To-Market: The Cold Start Strategy



▶ Day 1 Infrastructure: 3-5 EchoStar nodes cover the entire DMA.

The Catalyst:

Nielsen recently ceased monitoring in 137 small US markets, creating a verification vacuum. Blockchain PoD fills this void.

Before CMXS:

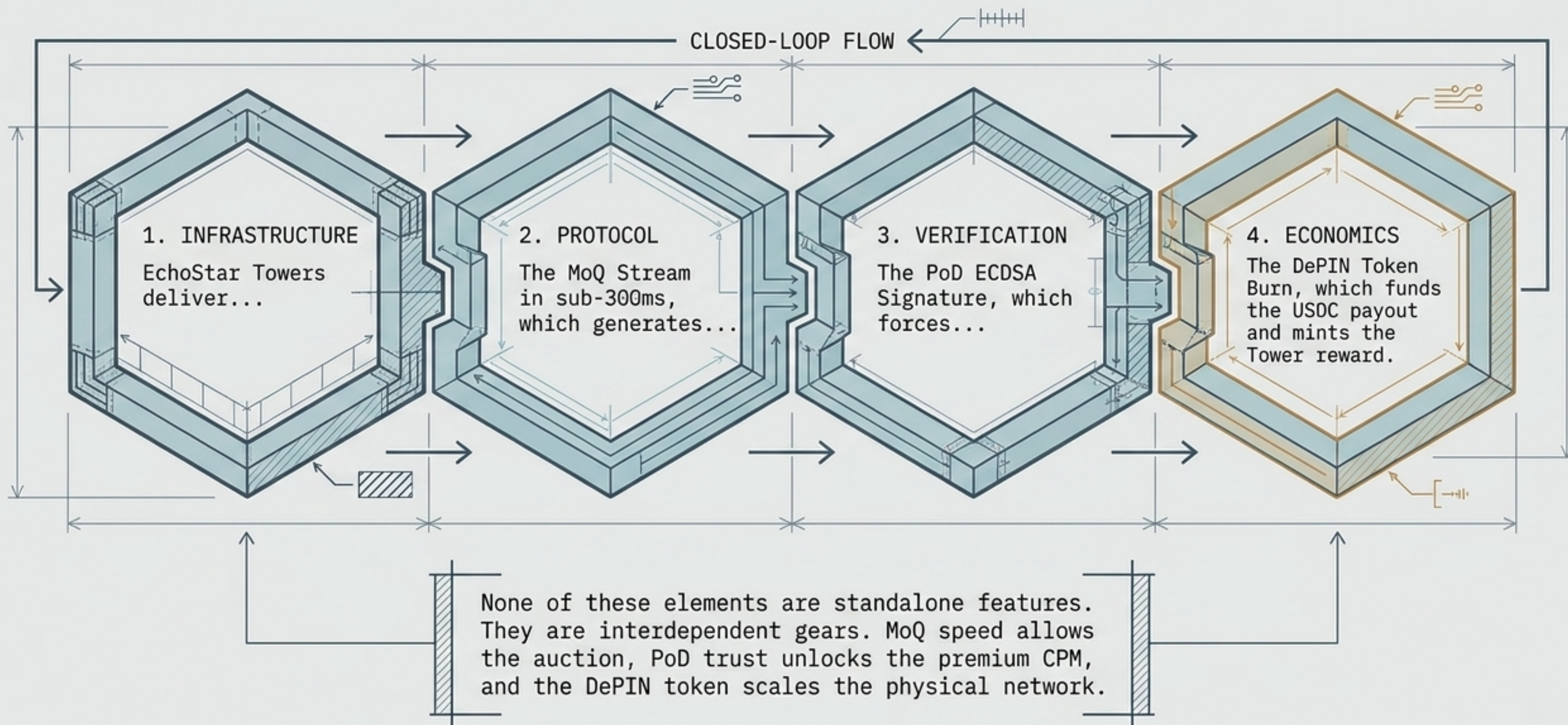
Local programmatic display ads yield \$1-\$5 CPM.

After CMXS:

Verified live local sports inventory yields \$38-\$45 CPM from regional advertisers (e.g., auto dealers).

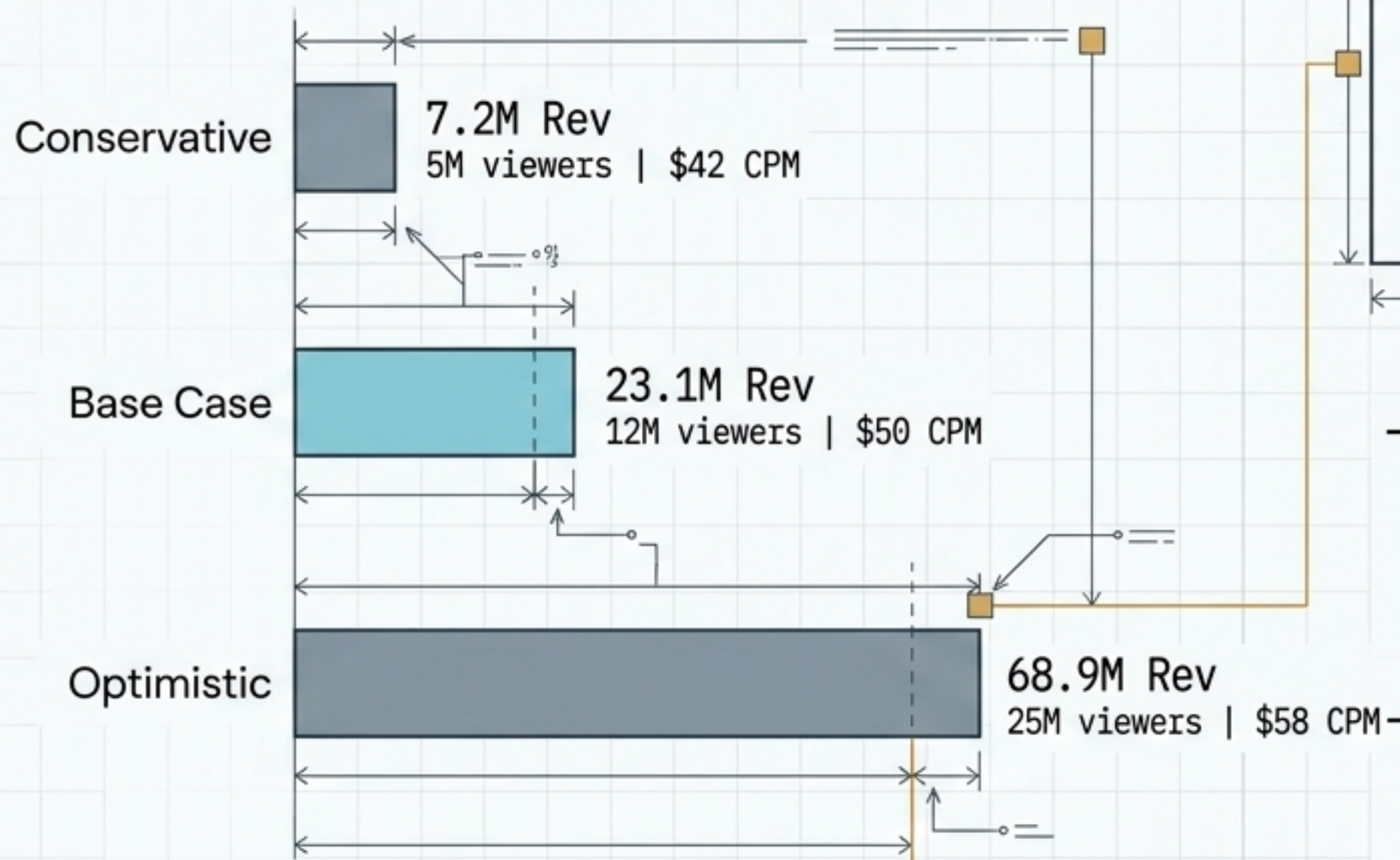
85% of gross verified CPM revenue goes directly to the local TV station. Zero capital expenditure required.

The Closed-Loop Economic Machine



Structural Value: Revenue Projections & Valuation Floor

Three-Scenario Revenue Model for Year 3



$$\frac{[\text{Annual Network Revenue} * \text{Revenue Multiple}]}{\text{Circulating Supply}} = \text{Rational Price Floor}$$

- Based on peer DePIN networks trading at 20-35x revenue multiples, CMXS sets an initial target FDV below what mainnet revenue justifies.
- The TGE is triggered only after auditable mainnet revenue is demonstrated on Basescan. Real revenue first. Token issuance second.